



Ch3_Metro Ethernet

Computer Networks

Opening ...

This chapter is

Objectives: learning...

- ◆ What is a MAN(Metro Area Network) and what topology does it have?
- ◆ MAN – Features, Technology, and Issues
- ◆ Metro Ethernet - Concept and Background
- ◆ Metro Ethernet - Features and Configurations
- ◆ What is required to deploy Metro Ethernet?

Contents



1. MAN Overview



2. Current MAN



3. Metro Ethernet



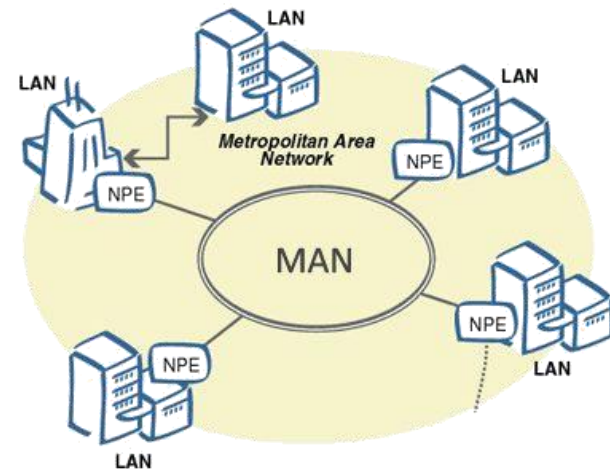
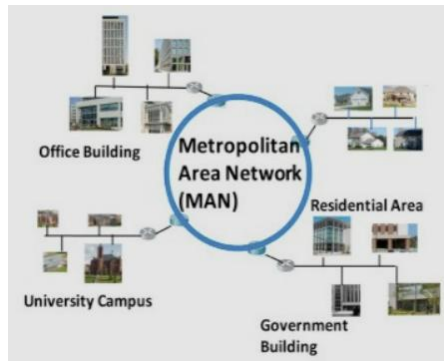
I. MAN Overview

1) Overview

MAN (Metropolitan Area Network) Overview

① MAN Overview

- **MAN: Metropolitan Area Network**
 - Connects subscriber networks (e.g., businesses and homes) across a metropolitan area
 - Provides connectivity between subscriber networks and the backbone network
- **Definition**
 - LAN-to-LAN connectivity
 - Provides a path to the backbone network



I. MAN Overview

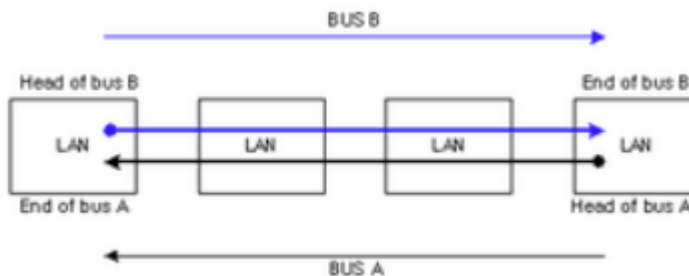
1) Overview

MAN (Metropolitan Area Network) Overview

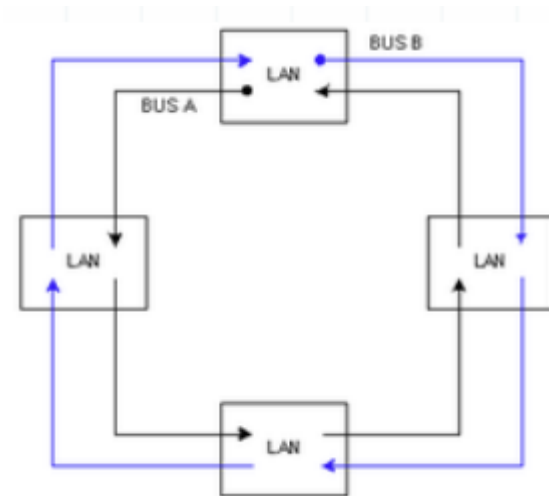
② MAN Topology

➤ DQDB(Distributed Queue Dual Bus)

- Facilitates communication between multiple LANs
- Uses a dual-bus structure
- Can be configured in a ring topology



(a) Dual Bus Topology



(b) Dual Ring Topology



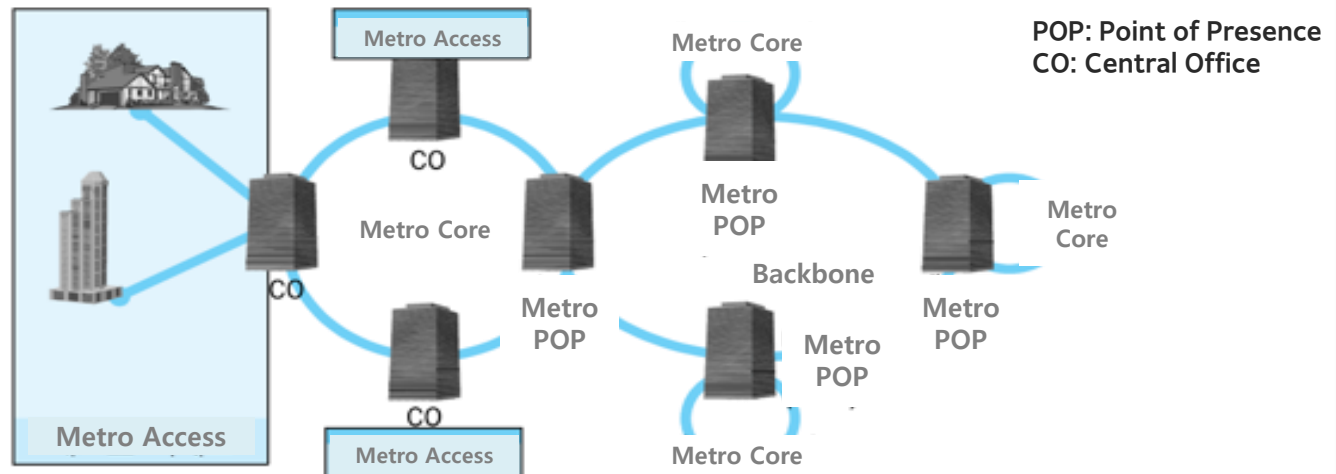
I. MAN Overview

1) Overview

MAN (Metropolitan Area Network) Overview

② MAN Topology

- A MAN typically uses a ring topology
 - A MAN consists of two main network segments:
 - Metro Access: Connects subscribers to the metro network
 - Metro Core: Connects metro access networks to the backbone



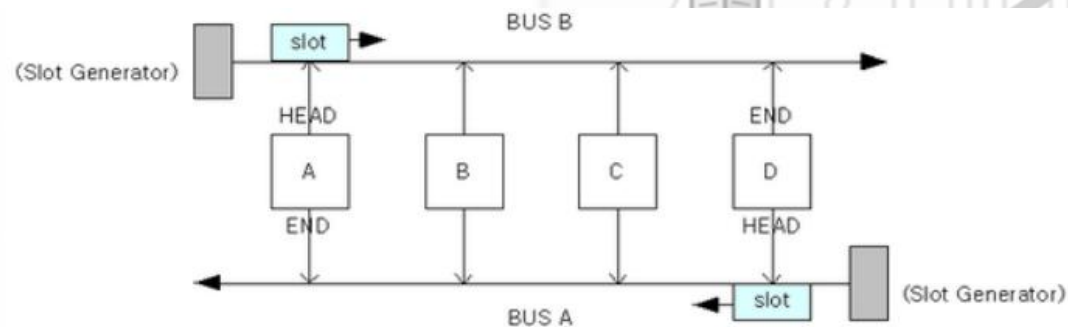
I. MAN Overview

1) Overview

MAN (Metropolitan Area Network) Overview

③ DQDB (Distributed Queue Dual Bus) - Legacy MAN Technology

- An early MAN technology standardized as IEEE 802.6
- Uses two unidirectional buses operating in opposite directions
- Uses fixed-size slots for data transmission
- Designed to interconnect LANs across metropolitan areas
- Historically supported data rates up to 155 Mbps
- Largely replaced by Ethernet-based MAN technologies



I. MAN Overview

1) Overview

MAN (Metropolitan Area Network) Overview

③ DQDB (Distributed Queue Dual Bus) - How DQDB Works

➤ Dual-Bus Structure

- Uses two unidirectional buses operating in opposite directions
- Each bus continuously carries fixed-size slots

➤ Slot-Based Transmission

- A node transmits data by using an available slot
- Data travels only in the direction of the selected bus

➤ Distributed Queue

- Nodes request transmission opportunities through the opposite bus
- Requests are managed as a distributed queue to provide fair access
- This mechanism helps prevent starvation

➤ Basic Operation

- Request → Wait for Turn → Find Available Slot → Transmit Data



I. MAN Overview

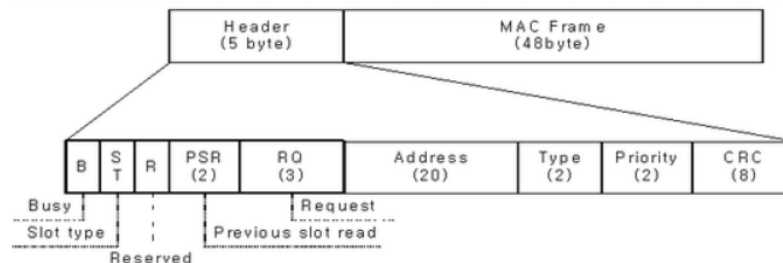
1) Overview

MAN (Metropolitan Area Network) Overview

③ DQDB (Distributed Queue Dual Bus)

➤ DQDB Frame

- B (Busy): Indicates whether the slot is occupied
- Slot Type (ST): Indicates the type of traffic carried by the slot (data or real-time traffic such as audio/video)
- Reservation (R): Reserved for future use
- PSR (Previous Slot Read): Indicates whether the slot was read by the previous node
- RQ (Request): Uses 3 bits to request a slot with a specific priority
- Address: Identifies the virtual channel (VCI)
- Type: Indicates whether the MAC frame carries data or control information
- Priority: Indicates the transmission priority
- CRC (Cyclic Redundancy Check): Detects transmission errors



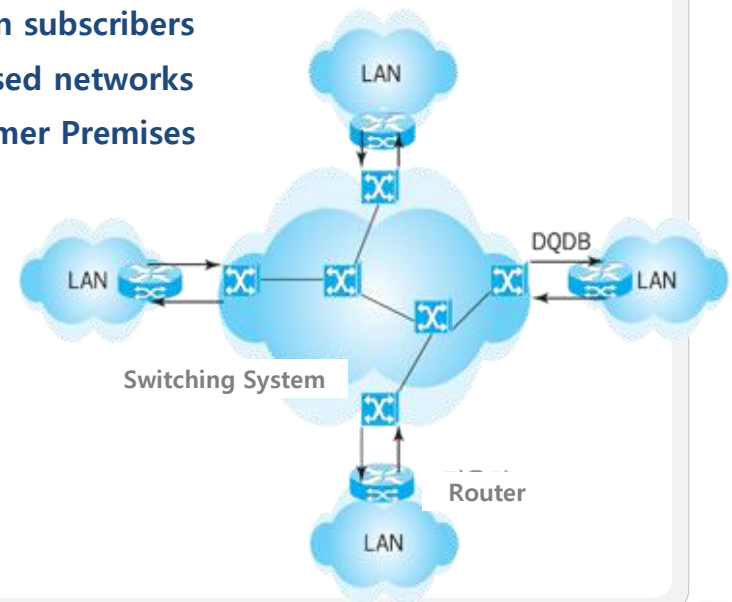
I. MAN Overview

1) Overview

MAN (Metropolitan Area Network) Overview

④ SMDS (Switched Multimegabit Data Services) - Legacy MAN Technology

- Designed for high-speed data communication over MANs and WANs
- Provides connectionless, packet-switched data services
- Supports data rates from 1.5 Mbps (T1) to 45 Mbps (T3)
- Uses network switches to efficiently forward data between subscribers
- Uses bandwidth more efficiently than traditional TDM-based networks
- Subscribers connect to the SMDS network through Customer Premises Equipment (CPE), such as a router



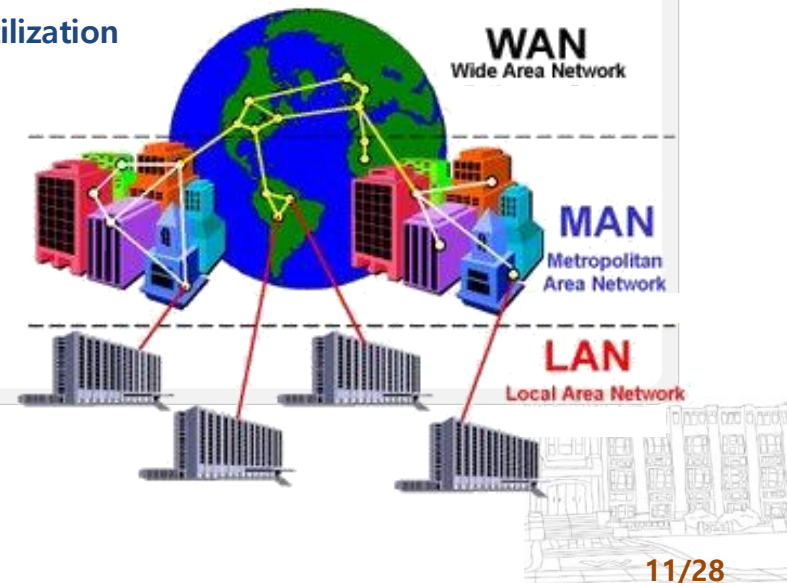
2. Current MAN

2) MAN

Current MAN (Metropolitan Area Network)

① Role of Current MAN

- Connects subscriber networks to the backbone network
 - Aggregates traffic from multiple subscriber networks
 - Provides high-speed connectivity within a metropolitan area
- Handles local traffic within the metropolitan area
 - Traffic between subscriber networks can remain within the MAN
 - Only necessary traffic is forwarded to the backbone
- Reduces traffic load on the backbone network
 - Improves overall network efficiency and backbone utilization



2. Current MAN

2) MAN

Current MAN (Metropolitan Area Network)

② MAN Configuration

➤ SONET/SDH Ring Architecture

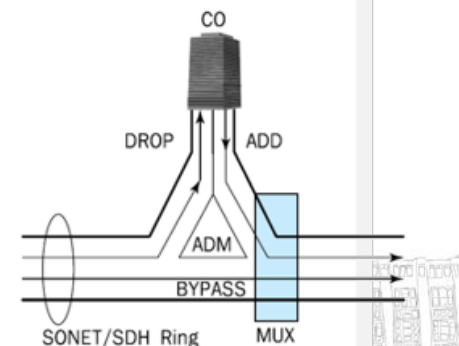
- Uses a high-speed optical ring to connect Central Offices (COs)
- Traditionally based on TDM for carrying voice and data traffic
- Provides reliable and high-capacity transport across a metropolitan area

➤ Add-Drop Multiplexer (ADM)

- Connects each CO to the SONET/SDH ring
- Add: Inserts local traffic from a CO into the ring
- Drop: Extracts traffic destined for a CO from the ring
- Bypass: Allows other traffic to pass through without processing

➤ Efficient Traffic Transport

- Each CO adds or drops only the traffic it needs
- Other traffic continues through the ring to its destination



2. Current MAN

2) MAN

Current MAN (Metropolitan Area Network)

② MAN Configuration

➤ Dual-Ring Architecture

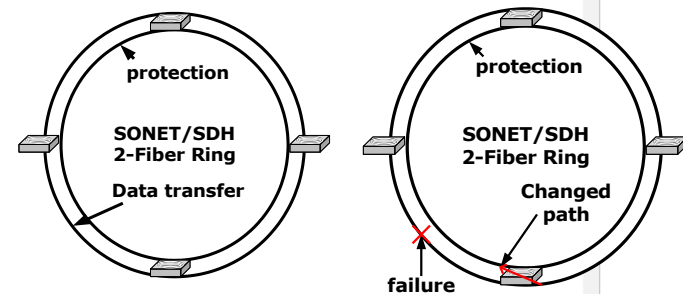
- SONET/SDH uses two fiber rings for reliable data transmission
- One path carries normal traffic, while the other provides a protection path
- The protection path can be used when the working path fails

➤ Fast Failure Recovery

- A link or node failure is quickly detected
- Traffic is automatically switched to the protection path
- SONET/SDH is designed to recover from failures within 50 ms

➤ High Network Reliability

- Ring protection minimizes service interruption
- Provides reliable transport for high-capacity MAN traffic



* Use protection link as data link



2. Current MAN

2) MAN

Current MAN (Metropolitan Area Network)

③ Problems with Traditional MANs

1. Inefficient Use of Bandwidth

- SONET/SDH was designed mainly for continuous voice traffic
- TDM reserves bandwidth even when no data is being sent

2. Rapid Growth of Subscriber Bandwidth

- Access networks became much faster
- MANs became a bottleneck between subscribers and the backbone

3. Increasing Metropolitan Traffic

- More servers and Internet services moved closer to users in large cities
- This greatly increased traffic within MANs

4. Growth of Multimedia Traffic

- Video, VoIP, and other multimedia services generated much more data traffic



2. Current MAN

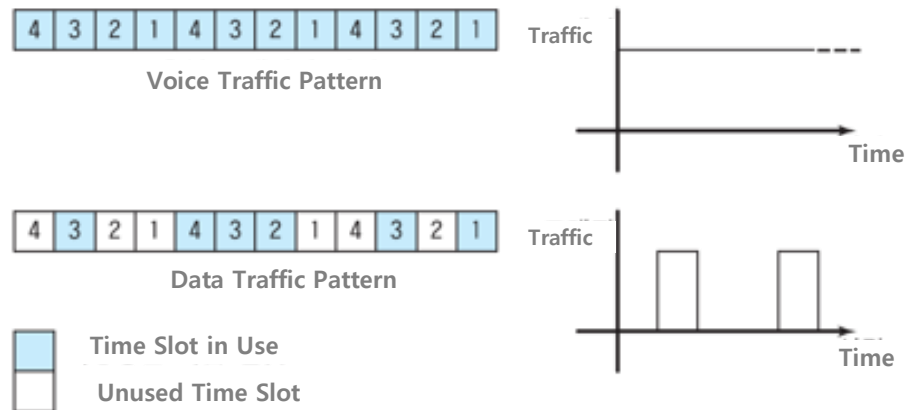
2) MAN

Current MAN (Metropolitan Area Network)

③ Problems with Traditional MANs

➤ Inefficient Bandwidth Use in TDM-based SONET/SDH

- Voice traffic is continuous, so reserved time slots are efficiently used
- Data traffic is bursty, with periods of activity and inactivity
- In TDM, time slots remain reserved even when no data is being transmitted
- As a result, network bandwidth can be wasted



2. Current MAN

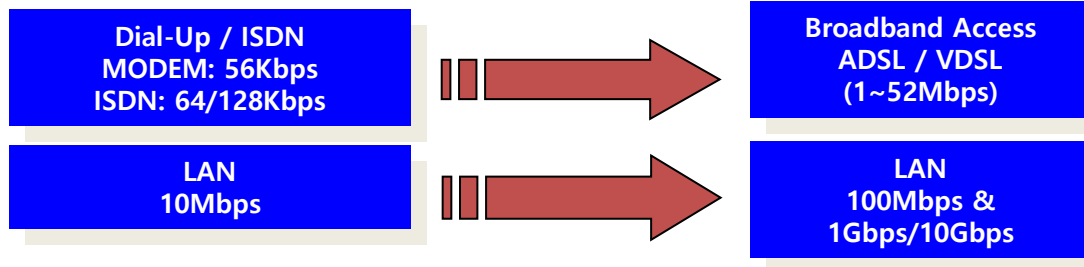
2) MAN

Current MAN (Metropolitan Area Network)

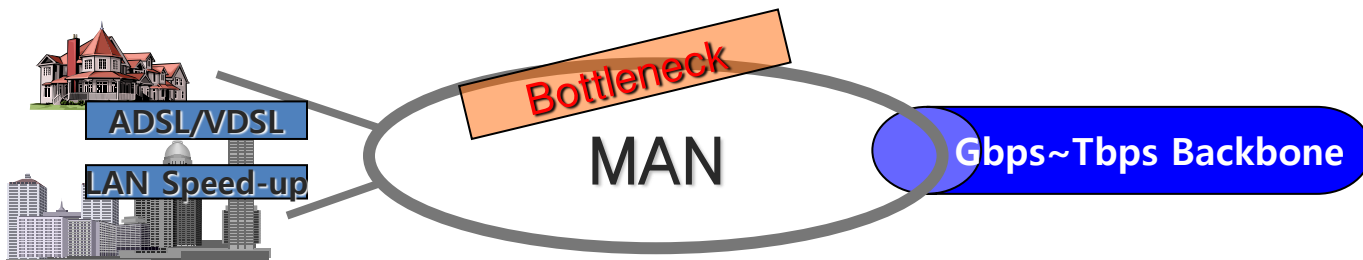
③ Problems with Traditional MANs

➤ Rapid Growth of Network Speed

- Subscriber Networks became much faster with broadband access and high-speed Ethernet



- Backbone networks also evolved to much higher speeds
- Traditional MANs could not keep up with this rapid traffic growth
- As a result, the MAN became a bottleneck between subscriber networks and the backbone



2. Current MAN

2) MAN

Current MAN (Metropolitan Area Network)

③ Problems with Traditional MANs

- Growth of Internet Services
 - Internet services became increasingly distributed across multiple locations
 - More servers and data centers were deployed closer to users
 - This increased traffic within metropolitan networks
- Growth of Multimedia Traffic
 - Video, VoIP, and other multimedia services generated large amounts of traffic
 - Streaming and on-demand services further increased bandwidth demand
- Result
 - Traditional MANs had difficulty handling the rapidly increasing traffic



3. Metro Ethernet

3) Metro Ethernet

Metro Ethernet

Metro Ethernet

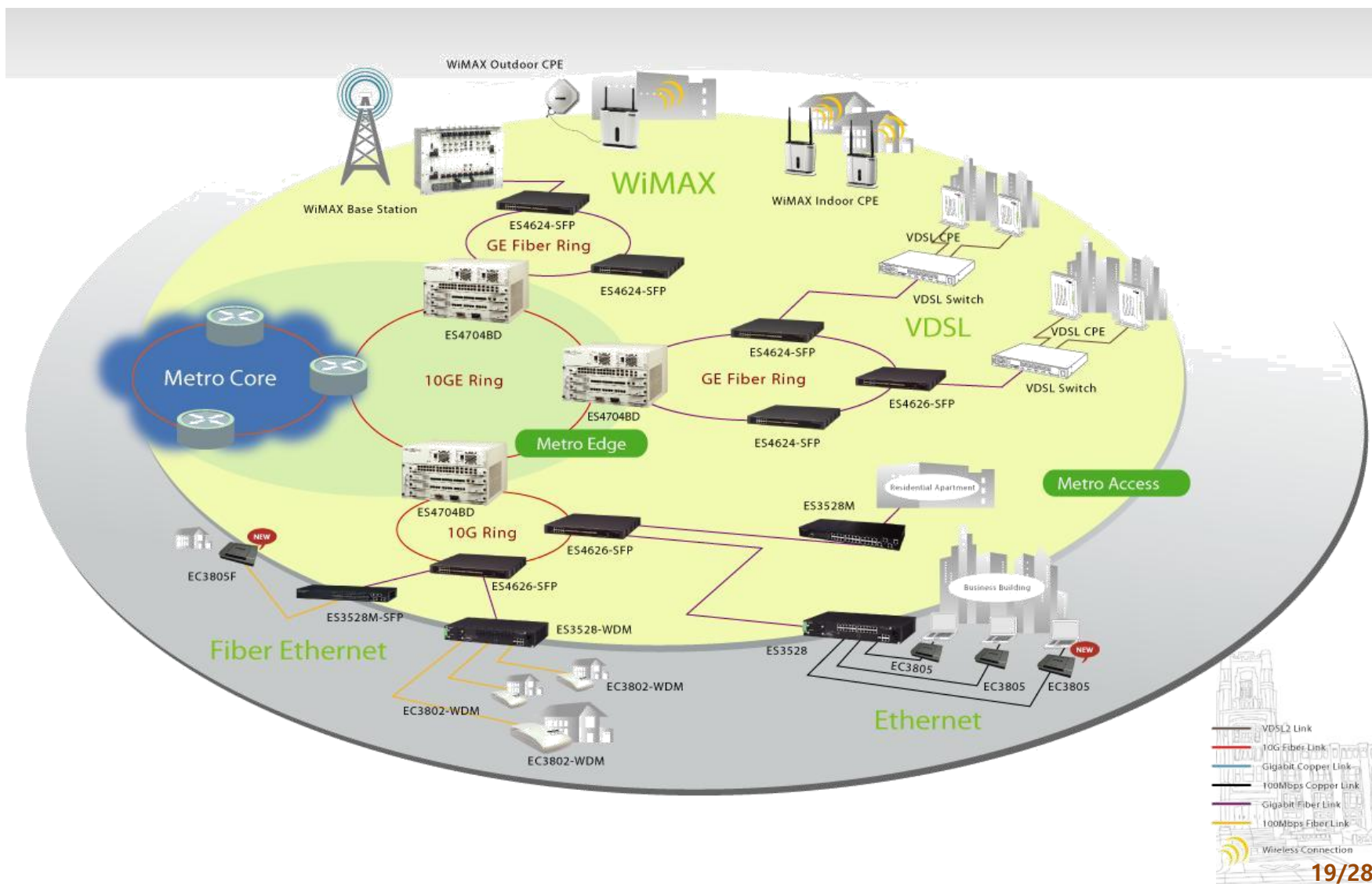
- What is Metro Ethernet?
 - Uses Ethernet technology to connect networks across a metropolitan area
 - Connects subscriber networks to service provider and backbone networks
 - Extends familiar LAN Ethernet technology to a city-wide network
- Why Metro Ethernet?
 - Uses packet switching, making bandwidth use more efficient than traditional TDM
 - Supports high-speed and scalable network connections
 - Easily connects Ethernet-based subscriber networks without complex protocol conversion
 - Helps overcome the bandwidth limitations of traditional MANs
- Physical Infrastructure
 - Typically uses high-speed optical fiber links
 - Can use available dark fiber to build high-capacity connections



3. Metro Ethernet

3) Metro Ethernet

Typical Metro Ethernet



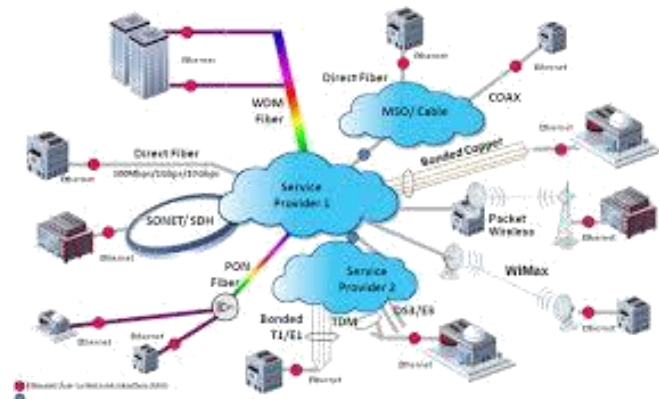
3. Metro Ethernet

3) Metro Ethernet

Metro Ethernet

① Metro Ethernet Features = Simple + Efficient + Scalable

- **Ethernet-Based Technology**
 - Uses widely adopted and well-established Ethernet technology
 - Supports high-speed connections and easy bandwidth upgrades
 - Uses widely available Ethernet equipment and expertise
 - Can evolve as Ethernet technology advances
- **Efficient Packet Switching**
 - Bandwidth is shared dynamically instead of being permanently reserved
 - Better suited to bursty data traffic than traditional TDM
 - Supports flexible bandwidth for different subscribers and services
- **Simple Network Integration**
 - Most subscriber LANs already use Ethernet
 - Reduces the need for protocol conversion between LANs and MANs
 - Ethernet technologies such as VLANs can be used across the metro network



3. Metro Ethernet

3) Metro Ethernet

Metro Ethernet

② Metro Ethernet Topology

➤ Mesh and Ring Topologies

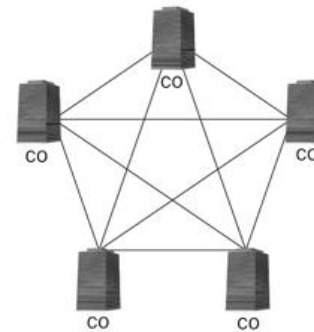
- Metro Ethernet can use either a mesh or ring topology
- Central Offices (COs) are connected using Layer 2 or Layer 3 switches

➤ Mesh Topology

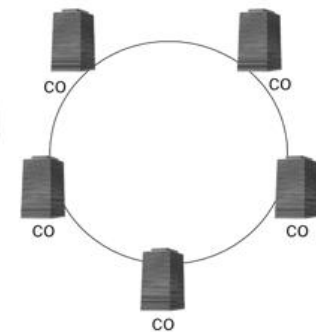
- Provides multiple paths between COs
- Full Mesh: Every CO is directly connected to every other CO
- Partial Mesh: Only selected COs are directly connected
- Provides high reliability but requires more links

➤ Ring Topology

- Connects COs in a ring structure
- Provides an alternate path when a link fails
- Requires fewer links than a full mesh



Mesh Topology



Ring Topology

3. Metro Ethernet

3) Metro Ethernet

Metro Ethernet

③ Metro Ethernet Function

➤ Packet Classification

- Classifies traffic according to service requirements
- Different traffic types can be assigned different priorities
 - Voice / Real-time traffic → High priority
 - Video traffic → Medium to high priority
 - Normal data traffic → Best effort

➤ How Traffic Is Classified

- Uses information in Ethernet, IP, and transport-layer headers
- IEEE 802.1p can provide priority information at the Ethernet layer
- DiffServ and MPLS can also be used for traffic classification and forwarding

➤ Why Packet Classification?

- Enables QoS (Quality of Service) for different types of traffic
- Helps control delay, bandwidth, and congestion

UP (User Priority)	WiFi Alliance	IEEE	
	AC (Access Priority)	802.1D/802.1p	802.1Q
1	AC_BK (Background)	BK (Background)	BK (Background)
2		-	EE (Excellent Effort)
0	AC_BE (Best Effort)	BE (Best Effort)	BE (Best Effort)
3		EE (Excellent Effort)	CA (Critical Application)
4	AC_VI (Video)	CL (Controlled Load)	VI (Video) < 100ms latency and jitter
5		VI (Video)	VO (Voice) < 10 ms latency and jitter
6	AC_VO (Voice)	VO (Voice)	IS (Internetwork Control)
7		NC (Network Control)	NC (Network Control)

lowest priority ↑

↓ highest priority

User Priority per each MSDU

Traffic Category

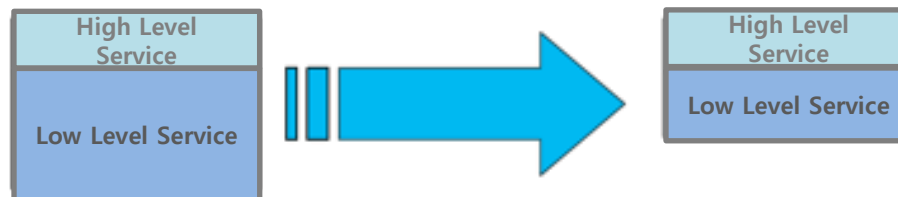
3. Metro Ethernet

3) Metro Ethernet

Metro Ethernet

③ Metro Ethernet Function

- **Ingress Traffic Control**
 - Controls the amount of traffic entering the Metro Ethernet network
 - Limits subscriber traffic according to the contracted bandwidth
 - Excess traffic may be delayed or dropped
- **Bandwidth Management**
 - Different service classes can receive different bandwidth guarantees
 - Packet classification is used to identify the service class
 - High-priority traffic receives guaranteed bandwidth when the network is congested
- **Purpose**
 - Prevents one subscriber from using excessive network resources
 - Helps provide predictable service quality to different subscribers and applications



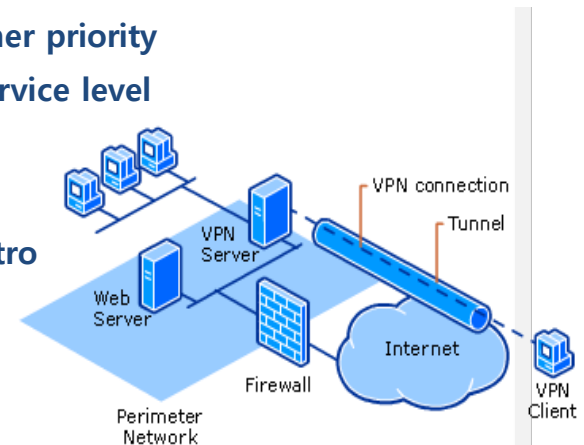
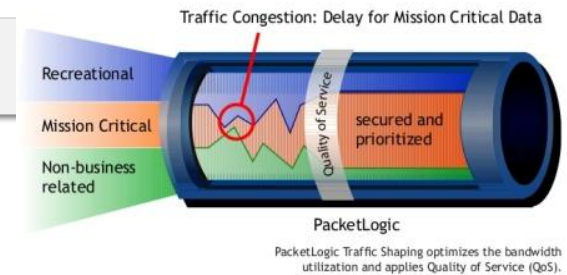
3. Metro Ethernet

3) Metro Ethernet

Metro Ethernet

③ Metro Ethernet Function

- **QoS (Quality of Service)**
 - Provides different service levels for different types of traffic
 - Traffic is classified and assigned different priorities
 - Delay-sensitive traffic, such as voice and video, can receive higher priority
 - Bandwidth can be guaranteed or controlled according to the service level
 - Helps maintain service quality when the network is congested
- **VPN (Virtual Private Network)**
 - Provides logically separated private networks over a shared Metro Ethernet infrastructure
 - Traffic from different customers is isolated from each other
- **How VPNs Are Created**
 - **VLAN:** Separates customer traffic using VLAN IDs
 - **MPLS:** Uses labels to separate and forward customer traffic across the provider network
 - VLAN and MPLS can be combined to support large-scale VPN services



3. Metro Ethernet

3) Metro Ethernet

Metro Ethernet

③ Metro Ethernet Function

- Network Resilience and Recovery
 - Metro Ethernet must continue operating even when a link or node fails
 - Redundant paths provide alternative routes for traffic
 - When a failure occurs, traffic is automatically switched to an available path
 - Ring-based networks can provide fast and efficient recovery
- Resilient Packet Ring (RPR)
 - RPR was developed to provide fast recovery in packet-based ring networks
 - Traffic can be redirected around the ring when a link fails



3. Metro Ethernet

3) Metro Ethernet

Metro Ethernet

③ Metro Ethernet Function

➤ Failure Recovery in RPR (Resilient Packet Ring)

- Wrap

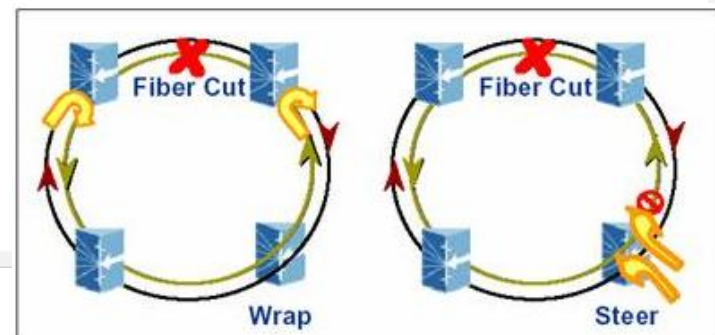
- Traffic is redirected at the nodes next to the failed link
- The traffic “wraps” onto the opposite ring to bypass the failure
- Provides fast recovery without changing the entire network path

- Steering

- Nodes are informed of the failed link
- The source node selects a new path that avoids the failure
- Traffic is sent directly through the available side of the ring

➤ Key Difference

- Wrap: Reroute traffic near the failure
- Steering: Select a new path from the source





Thank You !

Computer Networks